Guide for Non-Specialists: Policy-Perception Oscillations

This notebook serves as 'Notebook Zero' — a plain-English roadmap to the methodology and models used in this research project.

Important Project Safety Notice

Before executing or citing the findings in this notebook, please read the public guidance on what this project is and is not claiming:

docs/not_saying.md - *What This Theory Is NOT Claiming*

1. What This Project Is Investigating

In many policy domains, we observe that public policy and public sentiment swing back and forth like a pendulum. For example, standards-based accountability and test-based evaluation in education rose sharply under NCLB, triggered intense public backlash, and were subsequently rolled back under ESSA. This project asks: **Is this oscillation an unavoidable, cyclical property of policy system design?** Specifically, we model policy systems as containing **delayed negative feedback**: policy adjusts to public demand, but because legislation and implementation take time, the adjustment occurs with a lag. When feedback is delayed, corrections naturally overshoot, leading to periodic swings.

2. The Core Variables

Our model tracks five variables:

Variable	Math Symbol	What It Represents in the Real World	Example in Education Case
Public Demand	D_t	What direction the public wants policy to move.	Support for reform vs. retreat
Policy Position	P_t	The actual intensity/restrictiveness of policy.	Standardized testing frequency
Attention	A_t	How salient/visible the issue is to the media.	Media coverage of school testing
Backlash	B_t	Organized resistance or public anger.	Opt-out movements and petitions

Variable	Math Symbol	What It Represents in the Real World	Example in Education Case
Adaptive Norm	N_t	The baseline expectation of what is 'normal'.	Normalized historical testing levels

3. Pre-Registered Hypotheses & Falsification

A theory that can explain any result post-hoc is not scientific. We pre-register the following key checks:

- **Confirmation Rule (H_1)**: Policy intensity must Granger-cause backlash in our statistical models.
- **Confirmation Rule (H_2)**: States with longer policy implementation delays ($ au$) must exhibit larger oscillation amplitudes.
- **Falsification Control case**: We run our full analysis on a control case (Federal Highway Funding) where salience is low and benefits are immediate. If highway funding shows the same oscillatory swings as education, our theory is incorrect.